

**MELBOURNE SCORESBY, Australia**WMO: **958670**Lat: **37.8711S**Lon: **145.2561E**Elev: **262**StdP: **14.56**Time Zone: **10.00 (AUS)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 36.1       | 38.1 | 32.9                          | 27.7 | 48.0 | 34.5 | 29.5 | 47.2 | 22.2                  | 54.5 | 19.1 | 55.2 | 0.7                      | 40   | 0.459 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 21.7                   | 94.1            | 67.0 | 89.8 | 66.2 | 85.9 | 65.5 | 70.7                | 85.0 | 69.0 | 83.0 | 67.4 | 81.1 | 9.4                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 99.0 | 74.9 | 64.5 | 91.8 | 73.1 | 62.8 | 86.4 | 72.1 | 34.7          | 84.8 | 33.2 | 83.5 | 31.9 | 81.1 | 78.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 18.2          | 15.8 | 14.0 | DB                         | 32.1 | 102.6              | 1.4 | 4.9                                                | 31.1 | 106.1      | 30.2 | 109.0      | 29.4 | 111.7      | 28.4 | 115.2 | (4) |
| (5)               |      |      | WB                         | 31.4 | 73.7               | 1.7 | 1.6                                                | 30.2 | 74.8       | 29.3 | 75.7       | 28.3 | 76.6       | 27.1 | 77.7  | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 58.5   | 68.4  | 68.2  | 65.1  | 59.0  | 54.1  | 50.2  | 49.7  | 50.6  | 54.0  | 57.1  | 61.4  | 64.7 |
|                                                                                  | DBStd       | 8.80   | 7.31  | 6.70  | 6.64  | 5.58  | 4.70  | 3.87  | 3.56  | 4.15  | 5.13  | 6.05  | 6.81  | 7.15 |
|                                                                                  | HDD50       | 163    | 0     | 0     | 0     | 2     | 11    | 43    | 49    | 41    | 12    | 4     | 1     | 0    |
|                                                                                  | HDD65       | 2896   | 42    | 37    | 81    | 195   | 338   | 443   | 474   | 446   | 333   | 258   | 154   | 95   |
|                                                                                  | CDD50       | 3265   | 570   | 511   | 469   | 272   | 139   | 50    | 40    | 60    | 133   | 225   | 341   | 455  |
|                                                                                  | CDD65       | 524    | 147   | 127   | 85    | 16    | 1     | 0     | 0     | 0     | 4     | 14    | 45    | 85   |
|                                                                                  | CDH74       | 5988   | 1714  | 1349  | 886   | 140   | 2     | 0     | 0     | 0     | 16    | 200   | 578   | 1103 |
|                                                                                  | CDH80       | 2492   | 823   | 578   | 339   | 21    | 0     | 0     | 0     | 0     | 1     | 45    | 195   | 490  |
| Wind                                                                             | WSAvg       | 5.9    | 6.4   | 5.9   | 5.8   | 4.9   | 5.2   | 5.2   | 6.2   | 6.3   | 6.7   | 6.4   | 6.2   | 6.3  |
| Precipitation                                                                    | PrecAvg     | 35.2   | 2.4   | 2.4   | 2.5   | 3.1   | 2.8   | 3.3   | 2.9   | 3.1   | 3.5   | 3.4   | 3.0   | 2.8  |
|                                                                                  | PrecMax     | 46.9   | 4.5   | 7.0   | 4.3   | 7.9   | 5.7   | 6.6   | 5.5   | 4.7   | 6.8   | 6.6   | 6.5   | 6.7  |
|                                                                                  | PrecMin     | 22.0   | 0.5   | 0.2   | 1.3   | 1.0   | 0.7   | 0.9   | 1.2   | 1.0   | 1.3   | 1.1   | 1.1   | 0.3  |
|                                                                                  | PrecStd     | 6.3    | 1.1   | 1.8   | 0.8   | 1.6   | 1.3   | 1.3   | 1.1   | 1.1   | 1.5   | 1.5   | 1.4   | 1.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 103.5 | 96.7  | 95.3  | 82.7  | 71.9  | 64.3  | 62.7  | 68.3  | 75.9  | 85.3  | 90.3  | 98.8 |
|                                                                                  |             | MCWB   | 69.7  | 67.9  | 66.7  | 61.3  | 57.4  | 55.1  | 52.6  | 55.6  | 58.8  | 62.2  | 65.6  | 66.7 |
|                                                                                  | 2%          | DB     | 94.1  | 91.6  | 88.2  | 77.9  | 68.0  | 61.3  | 59.8  | 64.1  | 71.3  | 79.4  | 86.1  | 91.4 |
|                                                                                  |             | MCWB   | 67.2  | 66.7  | 65.1  | 60.9  | 57.0  | 53.3  | 51.5  | 52.9  | 57.6  | 60.4  | 64.1  | 65.6 |
|                                                                                  | 5%          | DB     | 89.1  | 87.3  | 83.2  | 74.3  | 65.2  | 59.1  | 57.7  | 61.2  | 67.6  | 74.7  | 81.0  | 85.3 |
|                                                                                  |             | MCWB   | 66.8  | 66.6  | 64.3  | 59.9  | 55.6  | 52.4  | 50.4  | 51.8  | 55.6  | 58.9  | 63.2  | 65.0 |
|                                                                                  | 10%         | DB     | 83.6  | 82.7  | 78.4  | 70.5  | 62.8  | 57.1  | 56.0  | 58.4  | 64.4  | 70.3  | 76.0  | 80.0 |
|                                                                                  |             | MCWB   | 66.2  | 65.7  | 63.4  | 58.8  | 54.6  | 51.3  | 49.5  | 50.4  | 54.0  | 57.2  | 61.7  | 63.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 73.3  | 72.7  | 69.9  | 65.0  | 60.3  | 56.5  | 54.7  | 56.7  | 61.2  | 65.1  | 69.9  | 71.0 |
|                                                                                  |             | MCDB   | 86.4  | 87.4  | 85.4  | 78.1  | 67.4  | 61.7  | 59.8  | 65.1  | 71.3  | 77.9  | 81.6  | 86.7 |
|                                                                                  | 2%          | WB     | 71.0  | 70.2  | 67.7  | 62.7  | 58.2  | 54.6  | 52.7  | 54.4  | 58.5  | 62.2  | 66.9  | 68.2 |
|                                                                                  |             | MCDB   | 86.1  | 83.5  | 81.0  | 72.9  | 65.1  | 59.3  | 57.7  | 62.2  | 69.0  | 74.7  | 79.2  | 83.1 |
|                                                                                  | 5%          | WB     | 69.1  | 68.4  | 66.1  | 61.1  | 56.7  | 53.2  | 51.3  | 52.5  | 56.6  | 60.2  | 64.9  | 66.4 |
|                                                                                  |             | MCDB   | 83.7  | 81.4  | 78.6  | 70.4  | 63.5  | 57.6  | 56.1  | 59.3  | 66.0  | 71.6  | 76.9  | 81.3 |
|                                                                                  | 10%         | WB     | 67.1  | 66.8  | 64.5  | 59.7  | 55.3  | 51.9  | 50.2  | 51.1  | 54.6  | 58.1  | 62.7  | 64.6 |
|                                                                                  |             | MCDB   | 81.8  | 79.0  | 76.2  | 68.9  | 61.8  | 56.4  | 54.8  | 57.2  | 63.3  | 68.9  | 74.1  | 77.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 21.7  | 21.0  | 19.7  | 18.0  | 13.7  | 12.2  | 11.6  | 13.2  | 15.5  | 18.8  | 19.8  | 21.1 |
|                                                                                  |             | MCDBR  | 32.6  | 30.2  | 28.7  | 24.6  | 17.3  | 15.3  | 13.4  | 17.3  | 20.6  | 27.9  | 29.7  | 32.7 |
|                                                                                  |             | MCWBR  | 10.6  | 10.3  | 10.1  | 10.8  | 9.2   | 9.3   | 8.0   | 9.1   | 10.3  | 12.3  | 11.8  | 11.6 |
|                                                                                  | 5% WB       | MCDBR  | 26.2  | 24.7  | 23.2  | 20.8  | 15.2  | 12.4  | 12.0  | 15.5  | 18.8  | 23.8  | 23.9  | 26.7 |
|                                                                                  |             | MCWBR  | 10.6  | 10.3  | 9.6   | 10.1  | 8.8   | 8.0   | 7.7   | 8.9   | 10.0  | 11.8  | 11.5  | 11.4 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.351  | 0.355 | 0.333 | 0.329 | 0.311 | 0.303 | 0.304 | 0.313 | 0.339 | 0.338 | 0.346 | 0.340 |      |
|                                                                                  | taud        | 2.505  | 2.500 | 2.562 | 2.546 | 2.583 | 2.593 | 2.580 | 2.544 | 2.463 | 2.490 | 2.493 | 2.531 |      |
|                                                                                  | Ebn at Noon | 311    | 302   | 297   | 278   | 265   | 258   | 265   | 279   | 290   | 305   | 311   | 316   |      |
|                                                                                  | Edh at Noon | 36     | 34    | 30    | 26    | 22    | 20    | 22    | 26    | 32    | 34    | 36    | 35    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2262   | 1974  | 1574  | 1091  | 715   | 556   | 620   | 879   | 1256  | 1690  | 1998  | 2231  |      |
|                                                                                  | RadStd      | 139    | 105   | 88    | 64    | 44    | 47    | 33    | 60    | 92    | 138   | 116   | 126   |      |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) |  | +0.92 | N/A     | N/A    | +1.53   | N/A   | N/A   | -32         | -229  | +311  | +106  |

Nomenclature: See separate page